CSIT751 (Core Java)

Module 4 – Collections and GUI programming

Programming Exercises for Practice (Practical Home Assignment 1)

1. ArrayList Basics

Problem Statement:
Write a Java program to create an ArrayList of strings, add five elements, and display them.

Sample Input Format:

Apple
Banana
Mango
Orange
Grapes

Sample Output Format:

ArrayList: [Apple, Banana, Mango, Orange, Grapes]

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2. Vector Usage

Problem Statement:
Write a Java program to store integers in a Vector, remove an element at a given index, and display the updated vector.

Sample Input Format:

10
20
30
40
50
remove index 2

Sample Output Format:

Updated Vector: [10, 20, 40, 50]

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3. HashSet Demonstration

Problem Statement:
Write a Java program to store unique integers in a HashSet and attempt to add duplicate values.

Sample Input Format:

10
20
30
20
40

Sample Output Format:

HashSet: [10, 20, 30, 40]

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4. LinkedList as Queue

Problem Statement:
Write a Java program to use LinkedList as a queue: enqueue elements, dequeue one element, and display the queue.

Sample Input Format:

1 2 3 4 5

Sample Output Format:

Queue after enqueue: [1, 2, 3, 4, 5]
After dequeue: [2, 3, 4, 5]

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5. HashMap Usage

Problem Statement:
Write a Java program to store key-value pairs (student name and marks) in a HashMap and retrieve marks for a given student.

Sample Input Format:

John 85
Alice 90

Bob 78
query Alice

Sample Output Format:

Marks for Alice: 90

6. TreeSet Sorting

Problem Statement:
Write a Java program to store integers in a TreeSet and display them in ascending order.

Sample Input Format:

5
1
4
2
3

Sample Output Format:

TreeSet: [1, 2, 3, 4, 5]

7. Hashing Example

Problem Statement:
Write a Java program to demonstrate hashing by displaying hash codes of different strings.

Sample Input Format:

Java
Python
C++

Sample Output Format:

HashCode of Java: 2301506
HashCode of Python: 75197497
HashCode of C++: 65123

8. Simple AWT Frame

Problem Statement:
Write a Java program to create an AWT Frame with a label 'Welcome to Java GUI'.

Sample Input Format:

No input required

Sample Output Format:

A window with label 'Welcome to Java GUI' is displayed

9. AWT Button Click Event

Problem Statement:
Write a Java program to create a Frame with a button. On clicking the button, display 'Button Clicked' on the console.

Sample Input Format:

Click the button

Sample Output Format:

Button Clicked

10. AWT Menu Example

Problem Statement:
Write a Java program to create a menu with 'File' and 'Edit' options.

Sample Input Format:

No input required

Sample Output Format:

A window with a menu bar containing File and Edit options is displayed

11. Swing JFrame

Problem Statement:
Write a Java program to create a Swing JFrame with title 'My Swing App'.

Sample Input Format:

No input required

Sample Output Format:

Swing window titled 'My Swing App' is displayed

12. Swing JButton and JLabel

Problem Statement:
Write a Java program to create a Swing window with a button and a label. Clicking the button changes the label text.

Sample Input Format:

Click the button

Sample Output Format:

Label text changes to 'Button Pressed'

13. GridLayout Example

Problem Statement:
Write a Java program to create a Swing window with 6 buttons arranged in a 2x3 grid layout.

Sample Input Format:

No input required

Sample Output Format:

Window displays 6 buttons in grid format

14. BorderLayout Example

Problem Statement:
Write a Java program to create a BorderLayout with components in North, South, East, West, and Center.

Sample Input Format:

No input required

Sample Output Format:

Window displays components in 5 regions

15. Event Handling - ActionListener

Problem Statement:
Write a Java program where clicking a button shows a message dialog saying 'Hello User'.

Sample Input Format:

Click the button

Sample Output Format:

Message dialog: Hello User

16. MouseListener Example

Problem Statement:
Write a Java program that changes the background color of a panel when the mouse enters and leaves it.

Sample Input Format:

Move mouse over panel

Sample Output Format:

Background changes on mouse enter and reverts on mouse exit

17. KeyListener Example

Problem Statement:
Write a Java program that displays the pressed key in a label.

Sample Input Format:

Press 'A' key

Sample Output Format:

Label: Key Pressed: A

18. Adapter Class Example

Problem Statement:
Write a Java program using WindowAdapter to close the application when the window is closed.

Sample Input Format:

Close the window

Sample Output Format:

Application terminates